# How-To: Lifecycle Hooks

ragent hooks let you run shell commands at key points in the session lifecycle. This guide covers the two mechanisms for controlling tool execution: **PreToolUse** and **PostToolUse** hooks.

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## Configuration

Hooks are defined in the `hooks` array of `ragent.json`:

```
{
  "hooks": [
    {
      "trigger": "pre_tool_use",
      "command": "/path/to/my-pre-tool-hook.sh",
      "timeout_secs": 30
    },
    {
      "trigger": "post_tool_use",
      "command": "/path/to/my-post-tool-hook.sh",
      "timeout_secs": 10
    }
  ]
}
```

Each hook entry has three fields:

| Field | Type | Default | Description |
|---|---|---|---|
| `trigger` | string | — | When to fire (see triggers) |
| `command` | string | — | Shell command to execute (runs via `sh -c`) |
| `timeout_secs` | integer | 30 | Timeout in seconds. Enforced for `post_tool_use` and the fire-and-forget triggers; **not** enforced for `pre_tool_use` (see PreToolUse Hooks). |

Invalid hook entries (for example an unknown `trigger` string) are skipped at load time with a log warning — no startup error surfaces to the user.

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## PreToolUse Hooks

PreToolUse hooks fire **before** a tool is executed. They can approve, deny, modify, block, or warn about the upcoming tool call.

Two mechanisms control tool execution, and they work side by side:

1. **JSON-Decision Protocol** — the hook writes a JSON decision to stdout.
2. **Exit-Code Convention** — the hook's exit code signals block / warn / error.

### JSON-Decision Protocol

When the hook exits with code **0**, ragent parses stdout as JSON. The following decisions are recognised:

| stdout JSON | Effect |
|---|---|
| `{"decision": "allow"}` | Skip the UI permission prompt and allow the tool. |
| `{"decision": "deny", "reason": "..."}` | Deny the tool call with an optional reason. |
| `{"modified_input": {"path": "new/path"}}` | Replace the tool's input arguments. |
| *(empty or invalid JSON)* | No decision — normal permission flow applies. |

**Example — allow a specific tool:**

```sh
#!/bin/sh
# Allow all read operations without prompting
echo '{"decision": "allow"}'
```

**Example — deny with a reason:**

```sh
#!/bin/sh
# Block writes to /etc
echo '{"decision": "deny", "reason": "Writes to /etc are not allowed"}'
```

**Example — modify input arguments:**

```sh
#!/bin/sh
# Redirect all file reads to a sandbox directory
echo '{"modified_input": {"path": "/sandbox/src/main.rs"}}'
```

### Exit-Code Convention

When the hook exits with a **non-zero** status code, ragent interprets the exit code **before** parsing stdout JSON:

| Exit code | Behaviour |
|---|---|
| **0** | Parse stdout JSON (see JSON-Decision Protocol). |

| Exit code | Behaviour |
|---|---|
| **1** | **Warn** — allow the tool to proceed, but emit `tracing::warn!` and publish `Event::HookWarning` on the event bus so the TUI can surface it. |
| **2** | **Block** — the tool call is blocked. The hook's stderr (trimmed, capped at 500 characters) is used as the block reason. Stdout JSON is **ignored**. |
| **≥ 3** | **Error** — treat as a hook failure (not a policy decision). Fall through to the normal permission flow with a `tracing::error!` diagnostic. |

**Example — block with exit code 2:**

```sh
#!/bin/sh
# Block all bash commands that contain "rm -rf"
if echo "$RAGENT_TOOL_INPUT" | grep -q "rm -rf"; then
  echo "Blocked: destructive command detected" >&2
  exit 2
fi
# Otherwise, allow without prompting
echo '{"decision": "allow"}'
```

**Example — warn with exit code 1:**

```sh
#!/bin/sh
# Warn when a write tool is used outside src/
if echo "$RAGENT_TOOL_INPUT" | grep -qv '"path":.*"src/'; then
  echo "Warning: write outside src/ directory" >&2
  exit 1
fi
exit 0
```

**Example — hook error with exit code 3:**

```sh
#!/bin/sh
# This hook has a bug (missing dependency)
if ! command -v jq >/dev/null 2>&1; then
  echo "Hook error: jq is not installed" >&2
  exit 3
fi
# Normal processing would go here
echo '{"decision": "allow"}'
```

When a hook exits with code ≥ 3, the tool call is **not** blocked — it falls through to the normal permission flow so a broken hook cannot lock the user out of all tool use.

### Precedence Rules

1. **Exit code 2 takes absolute precedence** over stdout JSON. Even if the hook writes `{"decision": "allow"}` to stdout, exit code 2 will block the tool call.
2. **Exit code 1 does not parse stdout JSON.** The warning is emitted regardless of stdout content.
3. **Multiple hooks are evaluated in order.** The first hook that returns a decision (`allow`, `deny`, `modified_input`, or `Blocked`) wins; remaining hooks are not executed.

4. **Spawn failure** is treated as exit code ≥ 3 (hook error). The tool call falls through to the normal permission flow.

> **No timeout on the pre-hook path:** PreToolUse hooks are spawned synchronously via `std::process::Command::output()` with no timeout wrapper, so `timeout_secs` is silently ignored here — a hanging pre-hook blocks every tool call until it exits. `timeout_secs` *is* enforced for `post_tool_use` hooks and the fire-and-forget triggers.

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## PostToolUse Hooks

PostToolUse hooks fire **after** a tool has executed. They can modify the tool's output, warn about the result, or flag it as policy-violated.

### Exit-Code Convention

| Exit code | Behaviour |
|---|---|
| **0** | Parse stdout JSON for `modified_output` (see below). |
| **1** | **Warn** — emit `tracing::warn!` and publish `Event::HookWarning`. The tool result is not modified. |
| **2** | **Flag** — publish `Event::ToolResultFlagged` with stderr as the reason. The tool result is **not** suppressed (the call already executed), but the flag appears in the session log and TUI. |
| **≥ 3** | **Error** — treat as a hook failure. No effect on the tool result. |

### Modified Output

When a PostToolUse hook exits with code 0 and writes JSON to stdout, ragent looks for a `modified_output` field:

```
{
  "modified_output": {
    "content": "Sanitised output here",
    "metadata": {"custom": "value"}
  }
}
```

If the `modified_output.content` field is present, it replaces the tool's output content.

**Example — redact secrets from tool output:**

```
#!/bin/sh
# Read the original output from the environment variable
OUTPUT="$RAGENT_TOOL_OUTPUT"
# Redact API keys (simplified example)
REDACTED=$(echo "$OUTPUT" | sed 's/sk-[a-zA-Z0-9]*/[REDACTED]/g')
# Return modified output
echo "{\"modified_output\": {\"content\": \"$REDACTED\"}}"
```

**Example — flag suspicious output:**

```sh
#!/bin/sh
# Check if the tool output contains sensitive data
if echo "$RAGENT_TOOL_OUTPUT" | grep -q "password"; then
  echo "Flagged: output contains potential password" >&2
  exit 2
fi
exit 0
```

---

## Other Hook Triggers

In addition to `pre_tool_use` and `post_tool_use`, ragent supports these triggers:

| Trigger | Description |
|---|---|
| `on_session_start` | Fired when a session receives its first user message. |
| `on_session_end` | Fired after a session completes processing a user message. |
| `on_error` | Fired when an LLM call or tool execution returns an error. |
| `on_permission_denied` | Fired when a tool call is rejected due to a permission rule. |
| `on_turn_start` | Fired at the start of each agent turn/iteration. |
| `on_turn_end` | Fired at the end of each agent turn/iteration. |
| `on_compaction` | Fired when context compaction is performed. |

These triggers fire asynchronously — errors are logged but never fatal. The `RAGENT_ERROR` environment variable is set for `on_error` hooks.

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## Environment Variables

All hooks receive these environment variables:

| Variable | Description |
|---|---|
| `RAGENT_TRIGGER` | The trigger name (e.g. `pre_tool_use`) |
| `RAGENT_WORKING_DIR` | The session working directory |

`on_error` hooks additionally receive:

| Variable | Description |
|---|---|
| `RAGENT_ERROR` | The error message |

PreToolUse hooks additionally receive:

| Variable | Description |
|---|---|
| RAGENT_TOOL_NAME | The name of the tool being invoked |
| RAGENT_TOOL_INPUT | JSON string of the tool arguments |

PostToolUse hooks additionally receive:

| Variable | Description |
|---|---|
| RAGENT_TOOL_NAME | The name of the tool that was invoked |
| RAGENT_TOOL_INPUT | JSON string of the tool arguments |
| RAGENT_TOOL_OUTPUT | JSON string of the tool output |
| RAGENT_TOOL_SUCCESS | "true" or "false" |

Turn and compaction triggers additionally receive:

| Variable | Description |
|---|---|
| RAGENT_TURN_NUMBER | Current agent turn/iteration number (on_turn_start, on_turn_end) |
| RAGENT_COMPACTION_REASON | Why compaction fired (on_compaction) |
| RAGENT_COMPACTION_TOKENS_BEFORE | Token count before compaction (on_compaction) |
| RAGENT_COMPACTION_TOKENS_AFTER | Token count after compaction (on_compaction) |

A **plugin-contributed** hook (see plugins.md section 9.6) additionally receives:

| Variable | Description |
|---|---|
| CLAUDE_PLUGIN_ROOT | The declaring plugin's root, so a command can resolve ${CLAUDE_PLUGIN_ROOT}/... |
| CLAUDE_PROJECT_DIR | Alias of RAGENT_WORKING_DIR, for Claude-dialect scripts |

and the event as JSON on **stdin** (the Claude hook protocol shape: hook_event_name, cwd, session_id, and — for the tool-scoped triggers — tool_name, tool_input, tool_response, tool_success). A hook declared in ragent.json keeps the environment-only contract (no stdin payload).

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## Stop Hook Continuation

The on_session_end trigger doubles as the Claude **Stop** hook. A Stop hook may block the turn and return findings (the Claude asyncRewake shape):

- exit 2 with a stdout JSON reason/additionalContext, else the stderr text;
- exit 0 with hookSpecificOutput.additionalContext; or
- a top-level {"decision": "block", ...}.

When a Stop hook returns non-empty findings they are appended to the conversation as a synthetic user turn (with an `AgentNotice`) **before the turn ends**, so a plugin security review can feed corrections back to the model. The loop is bounded to three continuations (`MAX_STOP_CONTINUATIONS = 3`), so a hook that always blocks cannot spin the session.

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## Complete Example

Here is a `ragent.json` snippet that configures both PreToolUse and PostToolUse hooks:

```
{
  "hooks": [
    {
      "trigger": "pre_tool_use",
      "command": ".ragent/hooks/pre-tool-use.sh",
      "timeout_secs": 10
    },
    {
      "trigger": "post_tool_use",
      "command": ".ragent/hooks/post-tool-use.sh",
      "timeout_secs": 10
    }
  ]
}
```

**.ragent/hooks/pre-tool-use.sh:**

```
#!/bin/sh
# PreToolUse hook: enforce security policies

TOOL="$RAGENT_TOOL_NAME"
INPUT="$RAGENT_TOOL_INPUT"

# Block bash commands that contain "rm -rf /"
if [ "$TOOL" = "bash" ] && echo "$INPUT" | grep -q "rm -rf /"; then
  echo "Blocked: refused to execute destructive command" >&2
  exit 2
fi

# Warn about writes outside the project directory
if [ "$TOOL" = "write" ] && ! echo "$INPUT" | grep -q '"path":.*"src/'; then
  echo "Warning: write target is outside src/" >&2
  exit 1
fi

# Allow all other tools without prompting
echo '{"decision": "allow"}'
```

**.ragent/hooks/post-tool-use.sh:**

```
#!/bin/sh
```

```
# PostToolUse hook: audit tool results

OUTPUT="$RAGENT_TOOL_OUTPUT"

# Flag outputs that contain potential secrets
if echo "$OUTPUT" | grep -qE "(sk-[a-zA-Z0-9]{20,}|AKIA[A-Z0-9]{16})"; then
  echo "Flagged: output contains potential API key" >&2
  exit 2
fi

# No modification needed
exit 0
```

---

## Summary

| Mechanism | PreToolUse | PostToolUse |
|---|---|---|
| Exit 0 | Parse stdout JSON for decision | Parse stdout JSON for `modified_output` |
| Exit 1 | Warn + allow (publish `HookWarning`) | Warn (publish `HookWarning`) |
| Exit 2 | Block (publish denial to agent) | Flag (publish `ToolResultFlagged`) |
| Exit ≥ 3 | Hook error → normal permission flow | Hook error → no effect on result |
| Spawn failure | Treated as exit ≥ 3 | Treated as exit ≥ 3 |
| Timeout | Not enforced (no timeout wrapper) | Treated as exit ≥ 3 (`timeout_secs`) |

When multiple PostToolUse hooks fire, the aggregated outcome follows the precedence **Flagged > Warn > Ok**, and among Ok results the last `modified_output` wins.

The two mechanisms (JSON decisions and exit codes) are **complementary**: JSON decisions provide fine-grained control (allow / deny / modify), while exit codes provide a simple, language-agnostic way to signal block, warn, or error from any scripting language.